



ADULT CONSENT FORM
PRINCETON UNIVERSITY

TITLE OF RESEARCH: *Predicting Algorithmic Trust at Scale*

PRINCIPAL INVESTIGATOR: Thomas Griffiths

PRINCIPAL INVESTIGATOR'S DEPARTMENT: *Psychology*

Key information about the study:

Your informed consent is being sought for research. Participation in the research is voluntary.

The U.S. Department of Defense is funding the study.

The purpose of the research: This project aims to collect data that can be used to evaluate when people will trust AI models, and create an effective method for measuring the extent to which humans can be expected to trust an AI system.

The expected duration of the subject's participation: The study duration will vary from 5 to 60 minutes.

The procedures that the subject will be asked to follow in the research and identification of any procedures that are experimental:

This study is administered in English. To participate, you should feel comfortable reading and writing in English. You will be presented with a task, then asked to make decisions based on some information. There will also be AI models making decisions about the task at hand, and you can choose to trust these models' responses or not. In some cases, you may also be asked to provide explanations or justifications for your responses, typically in the form of a short paragraph. Some tasks may involve emotionally sensitive or ethically complex scenarios. You may skip any questions you are not comfortable answering or withdraw at any time without penalty.

The tasks you will perform will be one or more of the following:

Risky/intertemporal choice: You will choose between two options. In the risky choice case, you are choosing between two lotteries where different monetary values occur with different probabilities. In the intertemporal choice case, you are choosing between different monetary values being received at different times.

Legal decisions: You will be asked to pick the legal response you trust more.

Moral/ethical decisions: You will be presented with a hypothetical scenario in which you or another entity faces a moral conundrum and has to decide which of two courses of action to pursue. You will be offered two choices and asked to choose one.

Text comprehension: You will read a passage of text and be asked to choose between two statements about the text, one of which is true and the other false.

Collaborative decision making: You will be shown the profiles of two candidates and choose which candidate to recruit for a collaborative task, based on the difficulty of the task and each candidate's competence and willingness to exert effort.

This study has been approved by the Princeton University IRB

Expert deference decisions: You will be presented with a hypothetical scenario in which you need to consult someone. You will be offered two options of people with different kinds of knowledge and asked to choose between those.

Multi-attribute choice: You will be shown choices between 2-4 options each defined by 2-4 attributes (e.g., three laptops defined by their price and speed) and asked to choose one.

Resource management: You will be shown “Purchase” vs. “Don’t Purchase” decisions for potential buying scenarios that vary in terms of price, payment type, source of funds, budget slack, bracketing, domain, sale frame, and sunk costs.

Pedagogical decisions: You will read a description of a student, a domain they are learning, and a problem that they are trying to solve. You will then be shown the student’s answer to the problem along with information about how the student solved the problem. The participant will then be presented with a pedagogical action to take in the scenario and will be asked to respond whether or not it is a good teaching action to take.

Image classification: You will be shown an image of a puzzle, such as a configuration of pieces in chess or movable objects in a grid and be asked to classify the puzzle as solvable or unsolvable. Additionally, you will be shown two images and asked to make a judgment of whether they are the same or different (e.g., the same object rotated).

Data interpretation: You will be presented with data and make a decision about which of two statements about those data are correct.

Aesthetic decisions: You will make a choice between 2-4 objects. The objects are either images, musical excerpts or stories. You will make an aesthetic decision (preference), indicating what you prefer.

Coding decisions: You will be presented with possible coding changes differing in levels of complexity and presentation. The presented changes will have potential bugs in them. You will be asked to select the best change among the ones presented.

Everyday decisions: You will be presented with text-based decisions between options in domains that include Health, Career, Relationships, Parenting & Caregiving, Finances, Education, Housing, Transportation, Lifestyle & Personal Development, and Legal & Civic settings.

You will also be asked to share demographic information (e.g. SES, education, native languages, etc.) and your experiences interacting with AI models. Some studies may ask you choose between deferring to another human or an AI model, choose between two different AI model responses, or ask whether you regret your choice to defer to an AI model.

For some study types, you may be rewarded or impeded on the basis of your judgments, decisions, or performance using pseudo-rewards such as money, points, and timeout penalties.

The reasonably foreseeable risks or discomforts to the subject as a result of participation:

Risks associated with participation in this study are minimal. You may feel slight discomfort answering some questions, or mild stress during time-constrained or evaluative tasks, but you may refrain from answering any questions that make you uncomfortable and may withdraw your participation at any time without penalty, even after completing the experiment. Worker IDs will be used to withdraw responses.

Note: During the experiment, you may read information generated by an AI model, and we want to remind you the risk of conversing with an AI language model/LLM includes potential misinformation from the model.

The benefits to the subject or to others, e.g., society that may reasonably be expected from the research: There are no direct benefits to you as a participant; however, by furthering our understanding of human cognition, this research will benefit society by helping with the development of automated systems that can better solve problems that are computationally challenging, but that people can solve easily with little formal guidance.

Confidentiality

This section should provide the following information:

What information will be collected?

We will be collecting survey responses. There will not be any identifiable information collected. The only identifier we collect will be your Worker ID which is pseudonymized. This allows us to link your responses to your Prolific account but not to your real identity. We collect this information because it is useful in verifying submissions, coordinating compensation, and re-contacting for additional studies. Data will only be stored digitally on password-protected devices and secure cloud systems. Any data shared will be anonymized by having Worker IDs removed.

Who will have access?

The research team members, and representatives of the U.S. Department of Defense will have access to research records as part of their responsibilities for human subjects protection oversight of the study.

How data will be protected?

Data will be stored on password protected devices and secure cloud systems. Secure transfer methods will be used for any data sharing across team members.

Limits of confidentiality?

This study does not have any limits to confidentiality.

How long data will be kept and what will happen to it afterward?

Data will be retained indefinitely and archived for future research. Future research will only have access to fully anonymized data. you may refrain from answering any questions that make you uncomfortable and may withdraw your participation at any time without penalty, even after completing the experiment. Worker IDs will be used to withdraw responses. Anonymized data may be shared in repositories.

Confidentiality protections for publications?

Findings will be reported in aggregate form and no individual will be identifiable in publications or presentations.

Voluntary Participation

You may refrain from answering any questions that make you uncomfortable and may withdraw your participation at any time without penalty, even after completing the experiment. Worker IDs will be used to withdraw responses.

Compensation

For your participation, you will receive [variable payment amount equivalent to \$10-\$15/hr]. If you are on MTurk/Prolific/Besample/CINT and for any reason you do not complete the study (e.g. technical difficulties, or a desire to stop), we will only be able to pay you if you send an email through MTurk/Prolific/Besample/CINT, or by emailing the researcher, at cocosci-lab@princeton.edu. If you have any questions about the study, feel free to contact the researcher or the Principal Investigator, Thomas Griffiths, at tomg@princeton.edu.

Who to contact with questions

Principal investigator: Dr. Thomas Griffiths, tomg@princeton.edu

If you have questions regarding your rights as a research subject, or if problems arise which you do not feel you can discuss with the Investigator, please contact the Institutional Review Board (IRB) at:

Phone: (609) 258-8543

Email: irb@princeton.edu

The IRB is an independent committee (not part of the research team) that reviews research to protect your rights and well-being.

Summary

I understand the information that was presented and that:

My participation is voluntary.

Refusal to participate will involve no penalty or loss of benefits to which I am otherwise entitled. I may discontinue participation at any time without penalty or loss of benefits.

I do not waive any legal rights or release Princeton University or its agents from liability for negligence.

By agreeing below, I confirm that I have read (or had read to me) and understand the information in this form, my questions have been answered, and I voluntarily agree to participate in this study.”